FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2019

PAPER – 2

TEST DATE: 28-04-2019

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

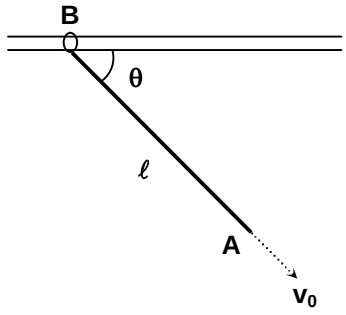
Physics

PART – I

SECTION – A
(One or More than one correct type)

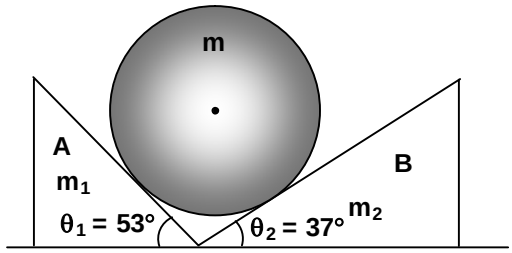
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A bead B of mass m = 0.5 kg which can slide along a smooth horizontal rod is attached to one end of a light inextensible string of length ℓ = 1m as shown. The other end 'A' of the string is pulled with a constant speed v_0 = 2 m/s always directed along the length of the string. Then choose the correct option(s).



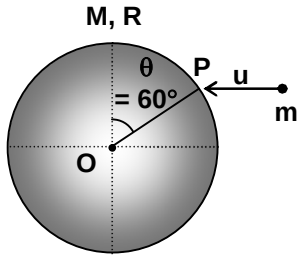
(A) The velocity of the bead at $\theta = 60°$ is 4 m/s.
(B) The velocity of the bead at $\theta = 60°$ is 1 m/s.
(C) The tension in the string at $\theta = 60°$ is 24 N.
(D) The tension in the string at $\theta = 60°$ is 12 N.

2. Two smooth triangular wedges A and B of masses m_1 = 9 kg and m_2 = 16 kg and inclination $\theta_1 = 53°$ and $\theta_2 = 37°$ respectively are placed on a smooth horizontal surface touching each other at their bottom edges. A horizontal cylinder of mass m = 25 kg is placed on the triangular wedges and then released from rest. When the centre of mass of the cylinder falls through height h = 0.4 m, then at this instant which of the following statements is/are correct.



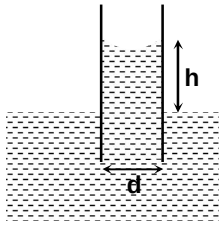
(A) The velocity of the wedge 'A' is 2 m/s.
(B) The velocity of the wedge 'B' is 1 m/s.
(C) The velocity of the cylinder is 1 m/s.
(D) The velocity of the cylinder is 2 m/s.

3. A uniform circular disc of mass M = 4 kg and radius R = 25 cm is suspended in a vertical plane and hinged at its centre 'O'. It is free to rotate about a horizontal axis passing through its centre and perpendicular to its plane. A particle of mass m = 0.5 kg moving horizontally with a velocity u = 20 m/s collides with disc at point 'P' and gets stuck to it as shown. Then choose the correct option(s). ($g = 10\ m/s^2$)



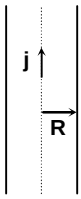
(A) The angular velocity of the disc just after collision is 4 rad/s.
(B) The angular velocity of the disc just after collision is 8 rad/s.
(C) The force on the disc due to hinge at its centre when the particle reaches the highest point is 38 N.
(D) The force on the disc due to hinge at its centre when the particle reaches the highest point is 44N.

4. Two vertical parallel rectangular glass plates are partially submerged in water. The separation between the plates is d = 0.1 mm and the width of each plate is ℓ = 10 cm. Assuming that the water between the plates does not reach the upper edges of the plates and the contact angle is θ = 0. The surface tension of water is T = 7×10^{-2} N/m and the density of water is $\rho = 10^3$ kg/m^3. Then choose the correct option(s). (g = 10 m/s^2)



(A) The water rises upto height h = 14 cm between the plates.
(B) The water rises upto height h = 28 cm between the plates.
(C) The force of mutual attraction between the plates is 4.9 N.
(D) The force of mutual attraction between the plates is 9.8 N.

5. A long current carrying cylindrical conductor of radius 'R' has a uniform current density 'j' over its cross section inside it. If we consider the two halves that are obtained by dividing the conductor by a plane containing the axis of the conductor. Then choose the correct option(s).



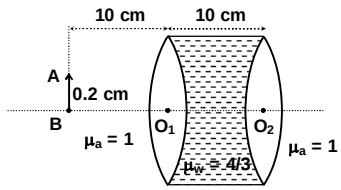
(A) The force of interaction per unit length between the two halves is $F = \left(\frac{\mu_0 j^2 R^3}{6}\right)$.

(B) The force of interaction per unit length between the two halves is $F = \left(\frac{\mu_0 j^2 R^3}{3}\right)$.

(C) The magnetic field inside the conductor at a radial distance $\frac{'R'}{2}$ from its axis is $B = \frac{\mu_0 jR}{4}$.

(D) The magnetic field inside the conductor at a radial distance $\frac{'R'}{2}$ from its axis is $B = \frac{\mu_0 jR}{2}$.

6. Consider an arrangement of two thin equiconvex lenses made of glass (μ_g = 3/2) and each of focal length in air 10 cm are separated by a distance 10 cm. The space between the two lenses is filled with water of refractive index μ_w = 4/3. A small linear object 'AB' of size 0.20 cm is placed perpendicular to the optic axis at a distance 10 cm from the first lens in air as shown. Then choose the correct option(s).

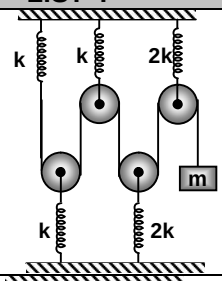
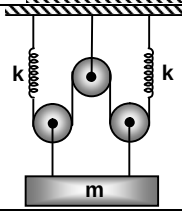
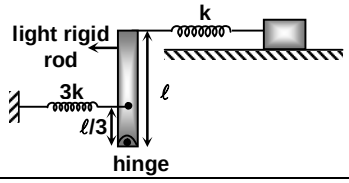
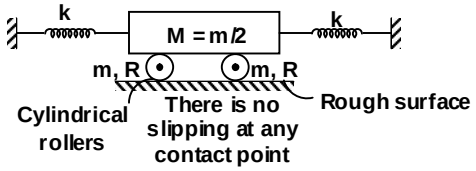


(A) The position of the final image formed is 25 cm right to the second lens.
(B) The position of the final image formed is 40 cm right to the second lens.
(C) The size of the final image formed is 0.40 cm.
(D) The size of the final image formed is 0.60 cm.

(Matching List Type)

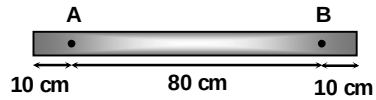
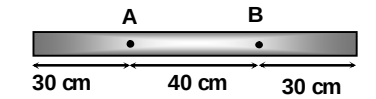
This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

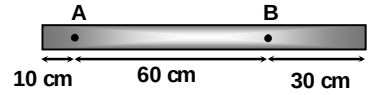
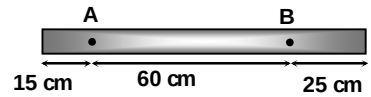
7. List –I represents the different spring mass systems. All springs and pulleys are ideal. List-II represents their time periods of small oscillations about equilibrium position. Match the systems mentioned in List-I with the corresponding time periods in List-II.

LIST–I		LIST–II	
P.		1.	$\pi\sqrt{\frac{m}{2k}}$
Q.		2.	$\pi\sqrt{\frac{5m}{2k}}$
R.		3.	$2\pi\sqrt{\frac{13m}{k}}$
S.		4.	$4\pi\sqrt{\frac{m}{k}}$

(A) P → 4; Q → 1; R → 3; S → 2
(B) P → 3; Q → 1; R → 4; S → 2
(C) P → 3; Q → 4; R → 1; S → 2
(D) P → 4; Q → 1; R → 2; S → 3

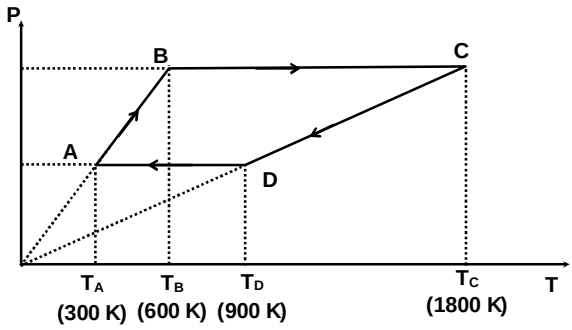
8. List – I represents the longitudinal vibration of metallic rods each of length 1 m and each rod is rigidly clamped at two points A and B. List – II represents their frequency and wavelength of longitudinal vibration in the fundamental mode of vibration. Then match the options of List-I correctly with the options of List-II.

LIST–I		LIST–II	
P.	 $Y = 2 \times 10^{11}$ N/m^2, ρ = 8000 kg/m^3	1.	f = 40 kHz
Q.	 $Y = 1.6 \times 10^{11}$ N/m^2, ρ = 2500 kg/m^3	2.	f = 12.5 kHz

R.	 $Y = 2 \times 10^{11}$ N/m^2, ρ = 8000 kg/m^3	3.	f = 20 kHz
S.	 $Y = 1.6 \times 10^{11}$ N/m^2, ρ = 2500 kg/m^3	4.	λ = 0.4 m
		5.	λ = 0.2 m

(A) P → 1, 5; Q → 3, 4; R → 2, 5; S → 1, 5
(B) P → 2, 5; Q → 3, 5; R → 2, 4; S → 2, 5
(C) P → 3, 4; Q → 3, 5; R → 2, 4; S → 2, 5
(D) P → 2, 4; Q → 3, 4; R → 2, 4; S → 1, 5

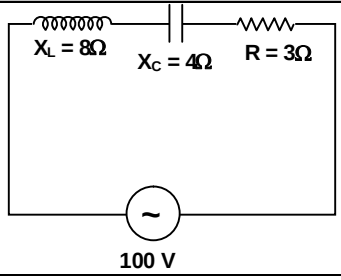
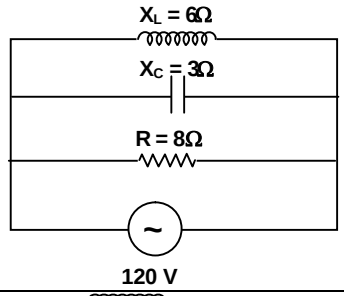
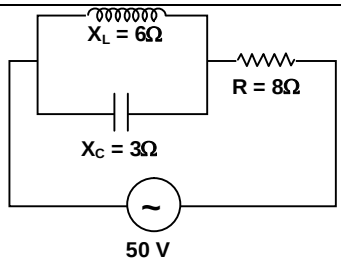
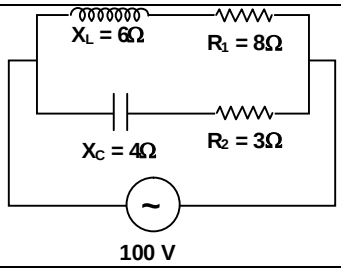
9. Two moles of an ideal monoatomic gas is taken through a cyclic process ABCDA as shown in the P-T diagram. List –I contains the four thermodynamic processes of the given cyclic process and List-II contains the quantities related to these thermodynamics processes. Match the options of List-I correctly with the options of List-II. (ΔQ = Heat supplied to the gas, ΔW = Work done by the gas)



LIST–I		LIST–II	
P.	In the process AB	1.	ΔQ = 900 R
Q.	In the process BC	2.	ΔQ < 0
R.	In the process CD	3.	ΔW = 2400 R
S.	In the process DA	4.	ΔW < 0
		5.	ΔW = 0

(A) P → 2, 5; Q → 3; 4; R → 2, 5; S → 2, 4
(B) P → 1, 5; Q → 3; 4; R → 2, 5; S → 2, 4
(C) P → 1, 5; Q → 3; R → 2, 5; S → 2, 4
(D) P → 2, 4; Q → 3, 4; R → 2, 4; S → 1, 5

10. List –I contains different AC circuits with rms voltage of ac sources as shown. List –II contains the corresponding rms currents supplied by the ac voltage sources. Then match the options of List-I correctly with the options of List-II.

LIST–I		LIST–II	
P.		1.	I_{rms} = 25 A
Q.		2.	I_{rms} = 5 A
R.		3.	$I_{rms} = 10\sqrt{5}$ A
S.		4.	I_{rms} = 20 A

(A) P → 4; Q → 1; R → 2; S → 3

(B) P → 2; Q → 3; R → 1; S → 4

(C) P → 3; Q → 2; R → 4; S → 1

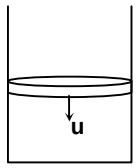
(D) P → 1; Q → 4; R → 2; S → 3

SECTION – D
(Numerical Answer Type)

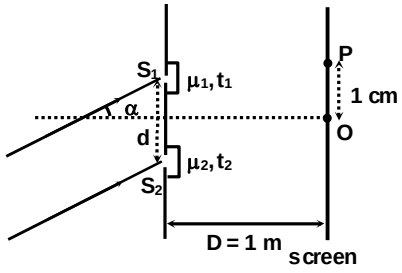
This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A string of mass 5 gram and length 50 cm is fixed at both ends. A pipe closed at one end has a length of 85 cm. When the string vibrates in its third overtone and the air column in the pipe in its first overtone, they produce a beats frequency of 4 Hz. It is observed that increasing the tension in the string increases beats frequency. Neglect the end correction in the pipe. The velocity of sound in air is 340 m/s. Find the tension (in Newton) in the string.

12. Two moles of an ideal monoatomic gas is enclosed in an adiabatic vertical cylinder fitted with a smooth adiabatic massive piston initially at rest. The whole system is placed inside a highly evacuated chamber. If the piston is imparted an initial velocity u = 8 m/s in vertical downward direction, after making a large number of oscillations, the piston finally comes to a complete stop. Find the displacement (in meter) of the piston. (Take g = 10 m/s^2)



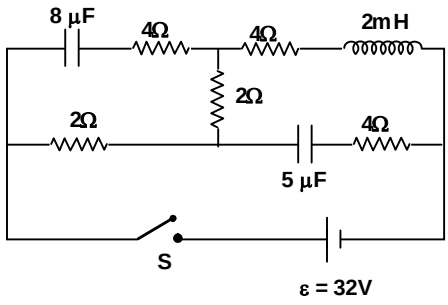
13. In young's double slit experiment two thin films are placed in front of two slits and the coherent parallel beam of light is incident at an angle $\alpha = 30°$ with the normal to the plane of slits as shown in the figure. If $\mu_2 = 1.8$, $\mu_1 = 1.5$ $t_2 = 3$ mm and the central maxima is formed at a distance 1 cm above point 'O' as shown. Find the thickness 't_1' (in mm) of the upper thin film.



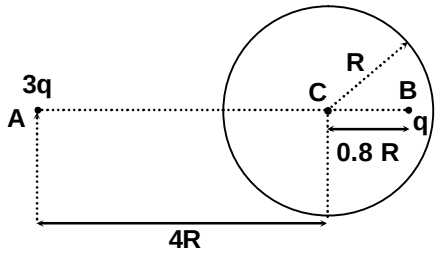
d = S_1S_2 = 2 mm

14. When a light of wavelength 5000 Å is incident on a metal surface of work function 1.9 eV, photoelectrons are emitted. When the emitted photoelectrons are passed through a region containing He^{2+} atoms, a fastest photoelectron combines with He^{2+} atom to form He^{+} atom and a photon is emitted. The He^{+} atom thus formed is in its third excited state. Then find the energy (in eV) of the photon emitted during the combination of the fastest photoelectron.

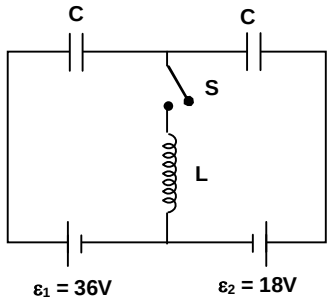
15. In the circuit shown, the switch 'S' is closed at t = 0. Then find the energy (in milli Joule) stored in 5μF capacitor in steady state.



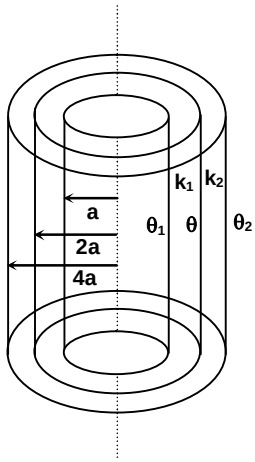
16. A thin conducting spherical shell of radius 'R' is given charge '2q'. Two point charges 'q' and '3q' are placed at distances 0.8 R and 4R from the centre 'C' of the shell respectively as shown. If the shell is earthed, find the charge (in μC) that will flow from conducting shell to the earth. (Given q = 2μC)



17. In the circuit shown, each capacitor has capacitance C = 10 μF and the inductor has inductance L = 2mH. The switch 'S' is opened for a long time. Find the maximum current (in ampere) through the inductor after the switch 'S' is closed.



18. A composite cylinder is made by joining two coaxial hollow cylinders as shown. The thermal conductivity of the material of the cylinder from radius 'a' to '2a' is K_1 = 2K and that for the cylinder from radius '2a' to '4a' is K_2 = 3K. The temperature of the innermost wall is maintained at θ_1 = 84°C and the outermost wall is maintained at temperature θ_2 = 32°C. Find the temperature θ (in degree Celsius) of the intermediate cylindrical wall of radius '2a'.

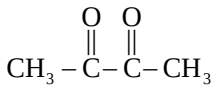


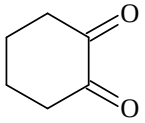
Chemistry PART – II

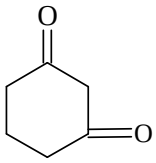
SECTION – A
(One or More than one correct type)

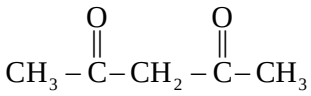
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. In which of the following cases enol content will be higher than the keto content in n-hexane?

(A) 

(B) 

(C) 

(D) 

20. In which of the following cases a mixture of optically active compounds with opposite optical rotation will not form when solutions are separated by semipermeable membrane
(A) 0.1 M D(+)–glucose & 0.1 M D(-) – fructose
(B) 0.2 M D(+)–glucose & 0.1 M D(-) –fructose
(C) 0.1 M D(+)–glucose & 0.2 M D(-) –fructose.
(D) 0.2 M D(+)–glucose & 0.2 M sugar

21. Which of the following compounds does not undergo complete hydrolysis?
(A) $SiCl_4$
(B) $BiCl_3$
(C) NF_3
(D) CCl_4

22. Which of the following statement(s) is/are correct ?
(A) $(SiH_3)_3N$ is a stronger base than $(CH_3)_3N$
(B) BF_3 is a weaker Lewis acid than BI_3
(C) Solid N_2O_5 is ionic in nature
(D) Phosphine is more basic than ammonia

23. Which of the following hydroxides dissolves in excess of NaOH?
(A) $Zn(OH)_2$
(B) $Al(OH)_3$
(C) $Sn(OH)_2$
(D) $Fe(OH)_3$

24. Decomposition of $3A(g) \rightarrow 2B(g) + 2C(g)$ follows 1st order kinetics. Initially only A is present in the container. Pressure developed after 20min and infinite time are 3.5 and 4 atm respectively. Which one is correct?

(A) $t_{50\%}$ = 20 min

(B) $t_{75\%}$ = 40 min

(C) $t_{99\%}$ = 64/3 min

(D) $t_{87.5\%}$ = 60 min

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match the following :

List I (Complex)		List II	
P.	$[NiCl_4]^{2-}$	1.	sp^3 hybridization
Q	$[Ni(CN)_4]^{2-}$	2.	dsp^2 hybridization
R	$Ni(CO)_4$	3.	Paramagnetic
S.	$[Pd(NH_3)_4]^{2+}$	4.	Diamagnetic
		5.	Tetrahedral geometry

(A) P $\rightarrow$ 1, 3, 5; Q $\rightarrow$ 2, 4; R $\rightarrow$ 1, 4, 5; S $\rightarrow$ 2, 4

(B) P $\rightarrow$ 3, 4; Q $\rightarrow$ 1, 4, 5; R $\rightarrow$ 1, 2, 3, 4, 5; S $\rightarrow$ 3, 5

(C) P $\rightarrow$ 2, 5; Q $\rightarrow$ 1, 4, 5; R $\rightarrow$ 1, 2, 3, 5; S $\rightarrow$ 3, 5

(D) P $\rightarrow$ 3, 5; Q $\rightarrow$ 1, 2, 3; R $\rightarrow$ 1, 2, 3, 4, 5; S $\rightarrow$ 2, 4

26. List -I have mixture of solutions and List – II shows pH of the resulting solution. Now, match the column.

	List – I		List – II
P.	A saturated solution of $Mg(OH)_2$ [$K_{sp}Mg(OH)_2 = 4 \times 10^{-12}$]	1.	pH = 7
Q.	300 ml of 0.1 M BOH + 100 ml. of 0.1 M HCl ($K_{b\,(BOH)} = 10^{-4}$)	2.	pH = $\frac{pk_{a_2} + pk_{a_3}}{2}$
R.	300 ml. of 0.1 M BOH + 300 ml. of 0.1 M HA [$K_a = K_b = 10^{-4}$]	3.	pH = 10 + log 2
S.	0.01 M aqueous solution Na_2HPO_4 [Given that : $pk_{a_1}(H_3PO_4) = 4.3$; $pk_{a_2}(H_3PO_4) = 8.102$; $pk_{a_3}(H_3PO_4) = 12.5$]	4.	pH = $\frac{1}{2} pK_w + \frac{1}{2} pk_a - \frac{1}{2} pk_b$
		5.	$[H^+] = 10^{-7}$

The correct option is

(A) P → 2; Q → 1, 2, 3; R → 1, 2, 3, 5; S → 2, 4

(B) P → 3; Q → 1, 4, 5; R → 1, 2, 3, 4, 5; S → 3, 5

(C) P → 2; Q → 1, 4, 5; R → 1, 2, 3, 5; S → 3, 5

(D) P → 3; Q → 3; R → 1, 4, 5; S → 2, 3

27. Match the following List I with List II

	List – I		List – II
P.	$N_{2(g)} + 3H_{2(g)} \Leftrightarrow 2NH_{3(g)}$; ΔH = –ve	1.	Low pressure favours the yield of the product
Q.	$N_{2(g)} + O_{2(g)} \Leftrightarrow 2NO_{(g)}$; ΔH = +ve	2.	Addition of He at constant pressure decreases the yield of the product
R.	$NH_4HS_{(s)} \Leftrightarrow NH_{3(g)} + H_2S_{(g)}$; ΔH= +ve	3.	Rate constant of forward reaction increases more than the increase of the rate constant of backward reaction with increase of temperature.
S.	$2NO_{2(g)} \Leftrightarrow N_2O_{4(g)}$; ΔH = –ve	4.	High temperature favours consumption of product.
		5.	Homogeneous equilibrium

The correct option is

(A) P → 2, 4, 5; Q → 3, 5; R → 1, 3; S → 2, 4, 5

(B) P → 3, 4; Q → 1, 4, 5; R → 1, 2, 3, 4; S → 3, 5

(C) P → 2, 5; Q → 1, 4, 5; R → 1, 2, 3, 5; S → 3, 5

(D) P → 3, 5; Q → 1, 2, 3; R → 1, 2, 4, 5; S → 2, 4

28. Match the following and choose the correct option.

	List – I		List – II
P.	PF_5	1.	Trigonal bipyramidal
Q.	PCl_5	2.	Zero dipole moment
R.	PH_5	3.	All atoms surrounding central atom are at same distance
S.	PCl_3F_2	4.	Solid state contains tetrahedral cation and octahedral anion
		5.	Does not exist

(A) P → 1, 2, 3; Q → 1, 4; R → 5; S → 1, 2, 3

(B) P → 1, 2, 3; Q → 1, 2, 4; R → 5; S → 1, 2

(C) P → 1, 3; Q → 1, 2, 5; R → 5; S → 1, 2, 3

(D) P → 2, 3; Q → 1, 2, 4, 5; R → 5; S → 1, 2, 3

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. 20 g of naphthoic acid ($C_{11}H_8O_2$) dissolved in 50 g of benzene(K_f=1.72 K kg mol^{-1}) shows a depression in freezing point of 2K. The Van't Hoff factor is

30. A fuel cell develops an electrical potential from the combustion of butane at 1 bar and 298 K
$C_4H_{10}(g) + 13/2\ O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(\ell)$ $\Delta G° = -2746$ KJ */mole*
The E°_{cell} of the fuel cell is

31. The equilibrium constant for the following reaction will be:
$HClO + CN^- \rightleftharpoons HCN + ClO^-$
Given Ka(HOCl)=3×10^{-8}
Ka(HCN) = 4.8 x 10^{-10}

32. Boiling point of an aqueous solution is 100.5°C. Given latent heat of fusion and latent heat of vapourisation of water are 80 cal g^{-1} and 540 cal g^{-1} respectively. The freezing point depression of the solution is nearly

33. Potassium dichromate in acidic medium reacts with H_2S to oxidize it as per given unbalanced equation. How much kg of $K_2Cr_2O_7$ must be required if 1000 kg of sulphur is to be produced by following reaction? (Molar mass of $K_2Cr_2O_7$ = 294 g/mol.)
$K_2Cr_2O_7 + H_2SO_4 + H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + S + H_2O$

34. 100 ml of 0.05 M $CuSO_4$(aq) solution was electrolysed using inert electrodes by passing current till the pH of the solution was 2. The solution after electrolysis was neutralized and then treated with excess of KI and the I_2 formed is titrated with 0.04 M $Na_2S_2O_3$. The volume of $Na_2S_2O_3$ required is

35. A steel vessel of volume one litre is filled with a mixture of CH_4 and O_2 in $1:2$ mole ratio at a total pressure of 1 atm at 27°C. The gas mixture is exploded for complete reaction. What would be the final pressure of the gaseous product at 127°C.

36. A certain acid–base indicator is red in acid solution and blue in basic solution. 75% of the indicator is present in the solution in its blue form at pH = 5, calculate the pH at which the indicator shows 90% red. [given log 3 = 0.477]

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If the roots of the equation $x^2 - px - 1 = 0$ and $x^2 - qx - 1 = 0$ form (in a suitable order) an arithmetic progression with four members then

(A) $p = \frac{2}{\sqrt{3}}, q = \frac{-2}{\sqrt{3}}$

(B) $p = \frac{4}{\sqrt{3}}, q = \frac{-4}{\sqrt{3}}$

(C) $p = \frac{-4}{\sqrt{3}}, q = \frac{4}{\sqrt{3}}$

(D) $p = \frac{-2}{\sqrt{3}}, q = \frac{2}{\sqrt{3}}$

38. For the curve $\sin x + \sin y = 1$ lying in the first quadrant $L = \lim_{x\to 0} x^{\alpha} \frac{d^2y}{dx^2}$, where L exists and is non zero. Then

(A) $\alpha = \frac{5}{2}$

(B) $\alpha = \frac{3}{2}$

(C) $L = \frac{1}{4\sqrt{2}}$

(D) $L = \frac{1}{2\sqrt{2}}$

39. Let M(–1, 2) and N(1, 4) be two points in a plane rectangular coordinate system XOY. P is a moving point on the x-axis. When $\angle MPN$ takes its maximum value, then which of the following is true

(A) P(–7, 0)

(B) Area of triangle PMN is 4 units

(C) Equation of circumcircle of triangle PMN is $(x + 7)^2 + (y - 10)^2 = 100$

(D) P(1, 0)

40. The plane $7x - 2y + z + 21 = 0$

(A) contains the point (–1, 7, 0)

(B) contains the line $\frac{x+2}{1} = \frac{3-y}{-2} = \frac{z+1}{-3}$

(C) is perpendicular to the line $\frac{x}{7} = \frac{y}{-2} = \frac{2}{1}$

(D) is parallel to the plane $7x - 2y + z + 5 = 0$

41. Given A(0, 2) and two point B and C on the parabola $y^2 = x + 4$ such that AB $\perp$ BC. Then the y-coordinate of point C may belongs to
(A) $y \in (-2, 2)$
(B) $y \in (-\infty, 0] \cup [4, \infty)$
(C) $y \in [5, \infty)$
(D) $y \in (-\infty, -3]$

42. If principal argument of z_0 satisfying $|z - 3| \le \sqrt{2}$ and $\arg(z - 5i) = -\frac{\pi}{4}$ simultaneously, is θ, then identify the correct statements
(A) $|z_0| = 17$
(B) $\tan 2\theta = \frac{8}{15}$
(C) $\tan\theta = -\frac{1}{4}$
(D) $|z_0 - 5i| = 4\sqrt{2}$

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. If $S_n = \sum_{r=1}^{n} r!$, then for n > 6 (where [.] denotes the greatest integer function)

$\alpha_1 = \sin^{-1}\left(\sin\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$; $\alpha_2 = \cos^{-1}\left(\cos\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$; $\alpha_3 = \tan^{-1}\tan\left(S_n - 7\left[\frac{S_n}{7}\right]\right)$;

$\alpha_4 = \cot^{-1}\cot\left(S_n - 7\left[\frac{S_n}{7}\right]\right)$, then which of the following is correct

LIST – I		LIST – II	
P.	Value of α_1	1.	$5 - \pi$
Q.	Value of α_2	2.	$5 - 2\pi$
R.	Value of α_3	3.	$6 - 2\pi$
S.	Value of α_4	4.	$2\pi - 5$

The correct option is:
(A) P → 2; Q → 4; R → 3; S → 1
(B) P → 2; Q → 4; R → 2; S → 1
(C) P → 4; Q → 2; R → 4; S → 1
(D) P → 4; Q → 2; R → 2; S → 3

44. Consider an ellipse (E) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, centred at point 'O' and having AB and CD as its major and minor axes respectively. If S_1 be one of its foci, radius of incircle of triangle OCS_1 be 1 unit and $OS_1 = 6$ unit

LIST – I		LIST – II	
P.	The length of the major axis of E is	1.	4
Q.	The maximum distance of normal from centre of E is	2.	$\frac{13}{2}$
R.	The distance between the directrix of E is	3.	$\frac{169}{12}$
S.	The length of the latus rectum of E is	4.	$\frac{25}{13}$

The correct option is:
(A) P → 2; Q → 1; R → 3; S → 4
(B) P → 1; Q → 2; R → 4; S → 3
(C) P → 2; Q → 1; R → 4; S → 3
(D) P → 1; Q → 3; R → 4; S → 2

45. N_1 be the number of zero's at the ends of $\prod_{m=5}^{30}(m)^{m+1}$.

N_2 be the number of three digit numbers of form xyz such that x < y and z ≤ y.

N_3 be the value of $\left[\frac{2009!+2006!}{2008!+2007!}\right]$, (where [.] denotes the greatest integer function).

N_4 be the number of positive integral pairs (x, y) satisfying the equation xy = 3370

LIST - I		LIST - II	
P.	The number of divisors of N_1	1.	8
Q.	The number of divisors of N_2	2.	12
R.	The number of divisors of N_3	3.	2
S.	The number of divisors of N_4	4.	4

The correct option is:
(A) P → 3; Q → 2; R → 1; S → 4
(B) P → 3; Q → 4; R → 1; S → 2
(C) P → 4; Q → 2; R → 3; S → 1
(D) P → 4; Q → 3; R → 2; S → 1

46. Each of the following function is defined to be zero at x = 0

$f_1(x) = x^2\,\text{sgn}(x)$

$f_2(x) = \int_0^x t^2 \sin\left(\frac{1}{t}\right)dt$

$f_3(x) = x^{-\frac{1}{3}}|\sin x|$

$f_4(x) = x^3[-x]$

where sgn(x) denotes signum function and [.] denotes the greatest integer function

LIST - I		LIST - II	
P.	The function f_1 is	1.	continuous but not differentiable at x = 0
Q.	The function f_2 is	2.	first derivative exists at x = 0 but second derivative does not exist
R.	The function f_3 is	3.	second derivative exists at x = 0, but it is not continuous there
S.	The function f_4 is	4.	second derivative exists at x = 0 and is continuous

The correct option is:
(A) P → 2; Q → 3; R → 4; S → 1
(B) P → 2; Q → 3; R → 1; S → 4
(C) P → 1; Q → 3; R → 4; S → 2
(D) P → 1; Q → 3; R → 2; S → 4

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. The real numbers α, β satisfy the equation $\alpha^3 - 3\alpha^2 + 5\alpha - 17 = 0$; $\beta^3 - 3\beta^2 + 5\beta + 11 = 0$. Then $\alpha + \beta$ is equal to

48. If $\log_4(x + 2y) + \log_4(x - 2y) = 1$. Then the minimum value of $|x| - |y|$ is $\sqrt{k}$. Then k is equal to

49. The value of $\tan 1^\circ \cdot \tan 2^\circ + \tan 2^\circ \cdot \tan 3^\circ + ... + \tan 88^\circ \cdot \tan 89^\circ$ is equal to $\cot^2 1^\circ - n$, then the value of n^2 is _____

50. Locus of a point P which divides all chords of slope $\frac{1}{2}$ of parabola $x^2 = 4y$ in the ratio 1 : 2 internally is another parabola with vertex $\left(\frac{a}{9}, \frac{b}{9}\right)$ and length of latus rectum $\frac{c}{9}$, then value of ab^2c^2 is

51. If $\sum_{k=1}^{\infty} \frac{k^2}{3^k} = p$, then the value of p is

52. A room contains a single bulb and $2^{2^{10}} + 2^{2^9}$ identical switches. A man can switch on or switch off any number of switches at a time. Find the minimum numbers of attempts required to identify the correct switch for that bulb (Initially all the switch are either on or off)

53. If $S = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} ..... \frac{1}{\sqrt{1000000}}$, then [S] is equal to
(where [.] denotes the greatest integer function)

54. Let S be the sum of digits of all the real coefficients of $(1 + ix)^{2015}$. If $\log_2 S = N$, then N is equal to